

School of Computing

Cisco Packet Tracer Tutorial

CSC1042 – Computer Networks II

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1. Introduction to Packet Tracer

Cisco Packet Tracer is a powerful, comprehensive network simulation tool designed to bridge the gap between theoretical learning and practical experience. Developed by Cisco Systems, it is widely used by students, educators, and network professionals to design, configure, troubleshoot, and simulate real-world computer networks without the need for physical equipment. Whether you're learning the basics of networking or preparing for advanced certifications, Packet Tracer offers an accessible platform to practice and refine networking skills.

With the growing complexity of modern networks, encompassing everything from small local area networks (LANs) to expansive wide area networks (WANs), there is a critical need for network engineers to have both theoretical knowledge and hands-on experience. Packet Tracer provides a virtual environment where users can experiment with network designs, configurations, and protocols, offering the flexibility to create everything from simple networks to intricate enterprise-scale topologies.

Why Cisco Packet Tracer is Essential

1. Cost-Effective Training Without Hardware: Building a physical network laboratory with routers, switches, and other networking devices is expensive and often impractical, especially for those learning network engineering at the introductory level. Cisco Packet Tracer eliminates the need for costly equipment by creating a simulated environment in which users can design and configure networks with realistic functionality. You can simulate real-world networks and interactions between devices, test configurations, and troubleshoot issues—all without needing access to physical devices.
2. Hands-On Learning with Realistic Configurations: Packet Tracer supports configuration and management via a Command-Line Interface (CLI) that mimics the actual interface found on Cisco devices. Whether you are configuring IP addresses, setting up routing protocols like OSPF or EIGRP, or troubleshooting network issues, Packet Tracer's CLI offers a near-identical experience to working with real Cisco hardware. This hands-on experience is critical for mastering network setup, management, and troubleshooting tasks, preparing you for Cisco certification exams such as CCNA and CCNP.
3. Interactive Simulation and Visualization: One of the standout features of Cisco Packet Tracer is its ability to simulate network traffic and protocol operations, providing insight into how data flows through a network. This is invaluable for learners, as you can visually observe how packets are forwarded between devices, how routing decisions are made, and how protocols like TCP/IP, ARP, or DNS function at the packet level. This ability to slow down network processes in Simulation Mode gives users a deeper understanding of network behaviour and protocol interactions.
4. Flexible Environment for Custom Scenarios: Packet Tracer is not limited to basic network setups; it is capable of simulating advanced scenarios, such as multi-tiered networks, VoIP, IoT (Internet of Things), and network security solutions. You can create setups to model real-world problems, such as simulating network failures and implementing network security measures like firewalls. This flexibility is key to preparing for real-life network management tasks, where network behaviour must be tested thoroughly before deployment.

5. Learning and Teaching Tool: Packet Tracer is widely used in educational settings as part of the Cisco Networking Academy curriculum. Its user-friendly interface, combined with robust functionality, allows students to experiment with network designs and configurations in a risk-free environment. Students can experiment with networks without the fear of "breaking" anything, making it an ideal learning tool.
6. Comprehensive Protocol Support: Modern networks rely on a wide array of protocols, and Cisco Packet Tracer supports many of them. Users can simulate and test protocols including:
    - Dynamic Routing Protocols: OSPF, EIGRP, RIP
    - Static and Default Routing
    - Switching: VLANs, STP (Spanning Tree Protocol)
    - IP Addressing: IPv4 and IPv6
    - Wireless Technologies: Wi-Fi with various security configurations
    - Security: firewalls, VPNs

By practicing with these protocols in Packet Tracer, you build a robust understanding of how networks are configured and how protocols interact to route and manage traffic.

2. The Importance of Physical and Logical Views

Cisco Packet Tracer has the ability to switch between Physical View and Logical View:

- **Logical View:** This is where you design and configure the network, showing a simplified diagram of how devices are connected and interact logically. It's the default workspace where you connect devices using cables, assign IP addresses, configure protocols, and verify connectivity. **Figure 1** shows an example of the logical view.

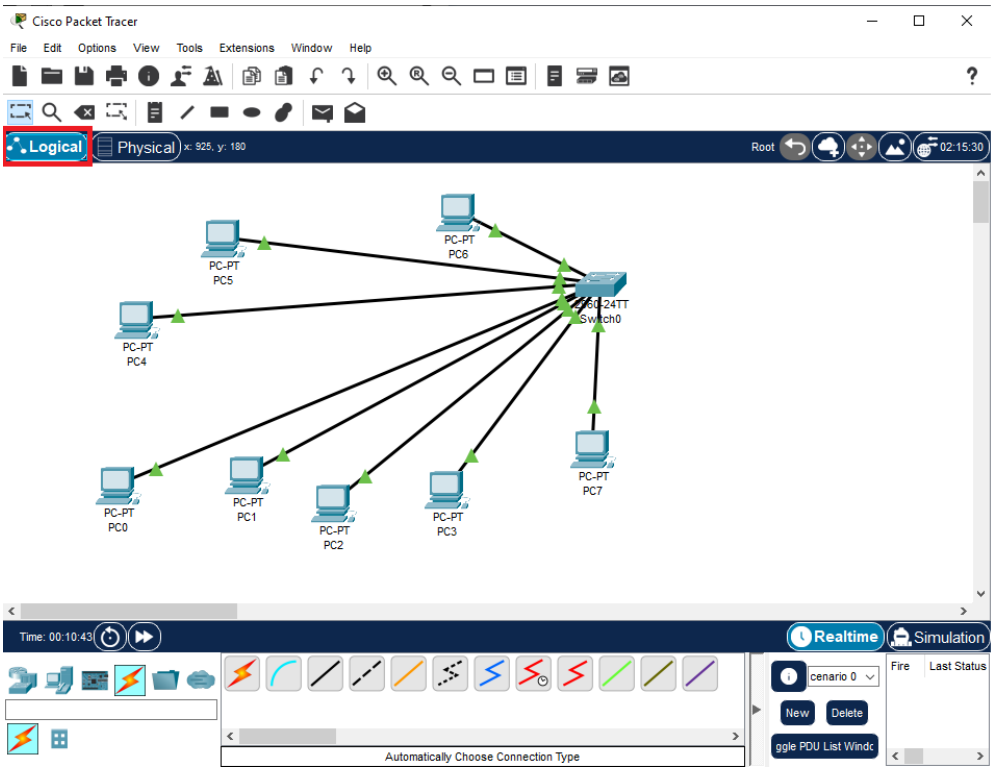


Figure 1: Example of a network setup shown in the logical view

- **Physical View:** This view shows the physical arrangement of the network, visualizing the physical layout of network devices, such as routers and switches, within racks and rooms. This mode is helpful for visualizing how a network might be physically deployed, such as a campus or data centre layout. **Figure 2** shows the same network in **Figure 1** in the real physical world depicting an office floor where PCs and the switch are placed in different rooms.

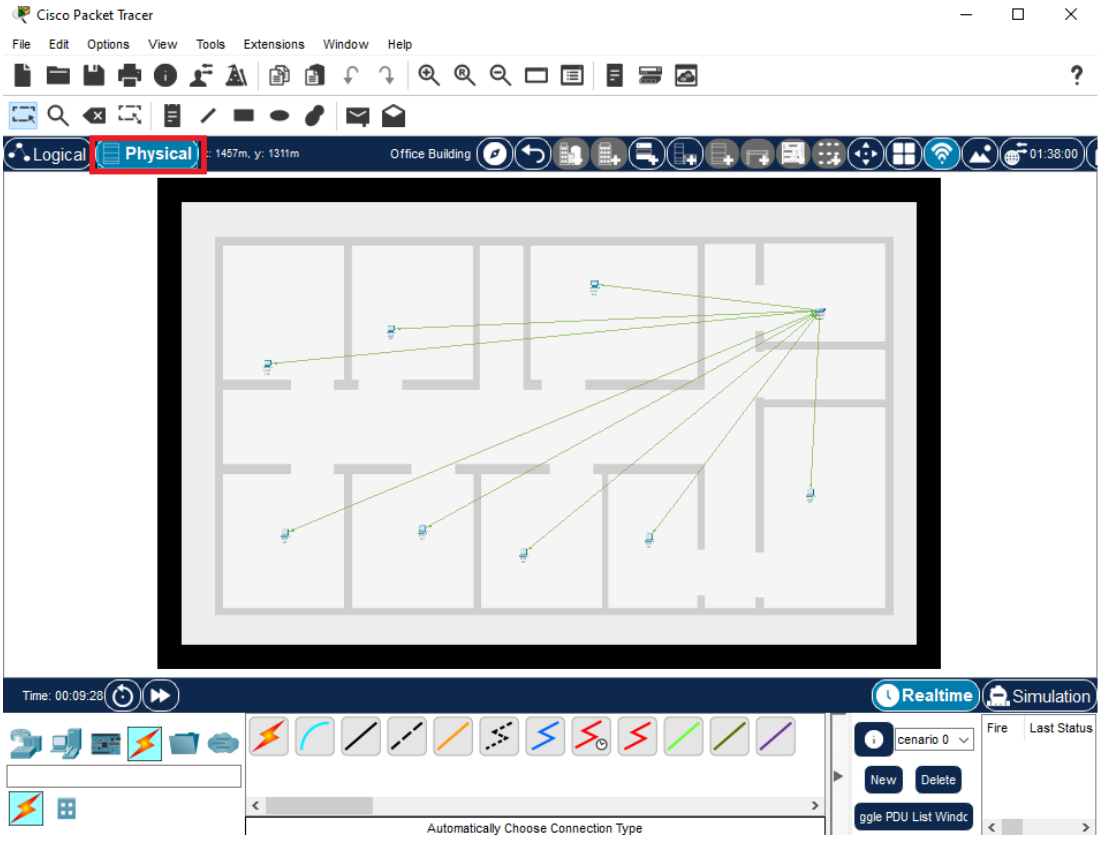


Figure 2: The same network setup is now shown in the physical view depicting an office floor where PCs and the switch are placed in different rooms

3. Real-Time and Simulation Mode: Understanding the Network's Behaviour

Cisco Packet Tracer offers two key modes to work with your network: ***Realtime Mode*** and ***Simulation Mode***. Both modes serve different purposes, making Packet Tracer a unique learning tool.

1. **Realtime Mode**: This is the default operating mode, where the network operates as it would in the real world. Changes to configurations, packet forwarding, and protocol behaviour happen instantaneously, just as they would on physical devices. You can test your network setup, deploy new configurations, and monitor how devices react in real time. **Figure 3** shows an example of network running under the ***Realtime*** mode.

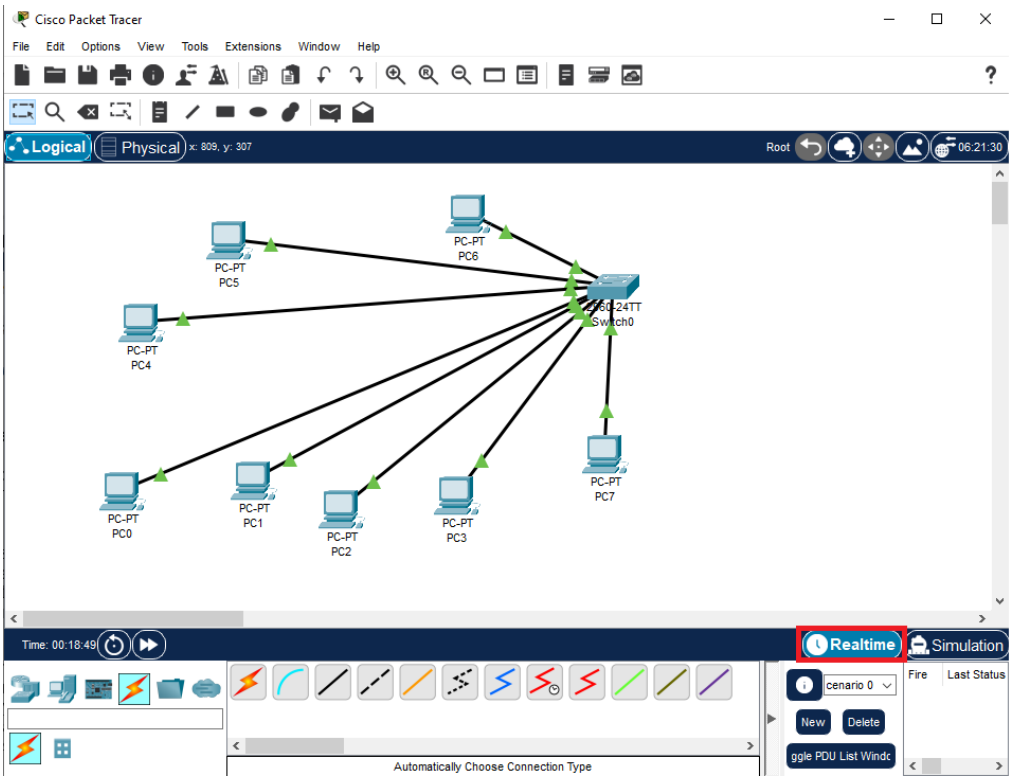


Figure 3: Example of a network running using the Realtime mode

2. **Simulation Mode**: For those who need to analyse the finer details of network traffic, Simulation Mode is invaluable. It allows you to slow down or stop time in the network, giving you the ability to trace packets as they traverse the network. You can monitor key protocols, such as ARP requests, DNS queries, TCP three-way handshakes, and ICMP responses, to fully understand how the network responds to different conditions. This mode is particularly useful when studying how protocols like OSPF or BGP handle network changes. **Figure 4** shows an example of a network running under the Simulation mode. Note that this mode allows you to track the types of messages exchanged between the various devices, which are displayed in the simulation panel (in blue). The mode also allows you to filter (in green) and display only the messages from protocols you are interested in. By default, all protocols are selected.

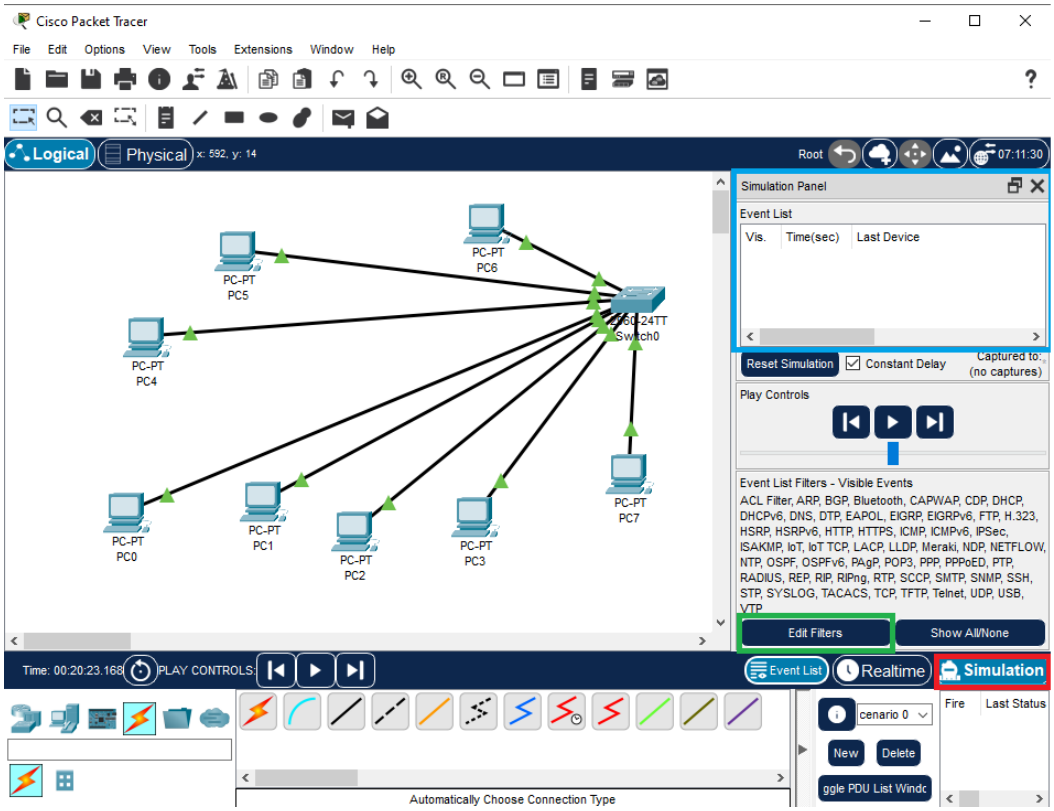


Figure 4: Example of a network running under the Simulation mode

4. Overview of Devices and Connections in Packet Tracer

Cisco Packet Tracer supports a wide variety of devices and connections, each with different capabilities. Here are the most commonly used devices:

End Devices:

- PC: A personal computer that can be used to simulate workstations in a network. PCs in Packet Tracer can be configured with IP addresses and used to ping other devices.
- Laptop: Similar to a PC but can connect wirelessly to networks.
- Smartphone/Tablets: These mobile devices can connect to wireless networks, allowing for IoT experimentation.
- Servers:
  - DHCP Server: Automatically assigns IP addresses to devices on the network.
  - DNS Server: Resolves domain names into IP addresses.
  - HTTP/FTP Server: Hosts web pages or file sharing services in the network.

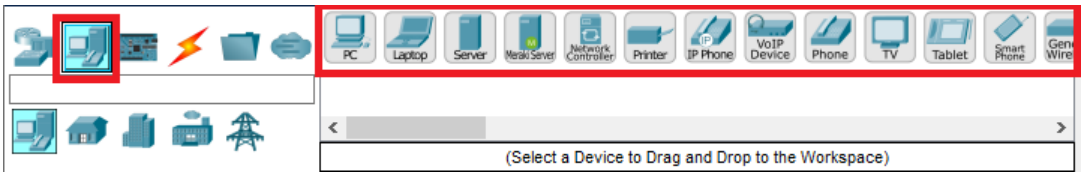


Figure 5: Types of end devices on Packet Tracer

Network Devices:

- Router: Used to route traffic between different networks. Routers are critical in connecting different subnets or WANs.
- Switch: A Layer 2 device that connects devices within the same network. Switches support VLANs, port security, and MAC address learning.
- Hub: A basic device that broadcasts data to all ports. Not used in modern networks due to inefficiency and lack of security.
- Wireless Router/Access Point: Provides wireless connectivity for devices, supporting Wi-Fi standards (e.g., 802.11).

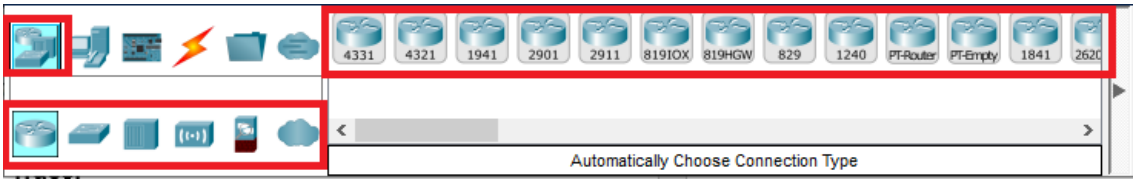


Figure 6: Types of network devices in Packet Tracer

Connections:

In Cisco Packet Tracer, simulating different network connections is essential for building and testing various network topologies. Here are the key types of connections you can simulate:

- Copper Straight-Through Cable: used to connect different types of devices (e.g., a switch to a router, or a PC to a switch).
- Copper Cross-Over Cable: used to connect similar devices (e.g., a switch to another switch, or a router to another router). It can also be used to connect a server to a router.
- Coaxial Cable: used to connect devices to a cable modem for Internet access.
- Fibre Optic Cable: used to connect devices over long distances or where high-speed data transmission is required.
- Serial DCE/DTE Cable: used to connect routers via WAN links, simulating connections through leased lines or dedicated long distance connections. One end is designated as DCE (Data Communication Equipment) and the other as DTE (Data Terminal Equipment).
- Console Cable: allows a computer to access a router or a switch's command line interface (CLI) for configuration or troubleshooting.

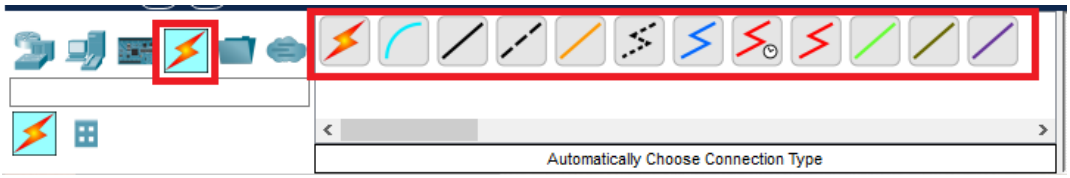


Figure 7: The connections types supported by Packet Tracer

5. Command-Line Interface Modes in Cisco Packet Tracer

CLI is a powerful tool in Cisco Packet Tracer that allows users to configure, manage, and troubleshoot Cisco devices like routers, switches, and firewalls. Cisco devices have a hierarchical CLI structure with different operational modes, each serving specific purposes and having different levels of access to the device's configuration.

Understanding the different CLI modes is crucial when configuring Cisco devices in Packet Tracer. Each mode allows you to perform specific tasks, from basic device checks to advanced configurations. Below is an overview of the key CLI modes and how to access and use them.

5.1 User EXEC Mode

The User EXEC mode is the default mode you enter when you first access the CLI of a Cisco device. It provides limited access and is primarily used for basic diagnostic commands, such as checking the device's status or verifying connectivity. Common commands include:

- `ping`: tests connectivity to other devices.
- `show version`: displays system hardware and software information.
- `show interfaces`: provides status information on interfaces.
- `exit`: Exits the CLI session.

You would know that you are in the User Exec Mode when you see “>” next to the device’s name, e.g., Router>

To display the list of commands under any mode, type ?

```
Router>?
Exec commands:
  <1-99>      Session number to resume
  connect     Open a terminal connection
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  ping        Send echo messages
  resume      Resume an active network connection
  show        Show running system information
  ssh         Open a secure shell client connection
  telnet      Open a telnet connection
  terminal    Set terminal line parameters
  traceroute  Trace route to destination
Router>
```

Figure 8: List of commands under the User EXEC Mode in a router

5.2 Privileged EXEC Mode

Privileged EXEC mode gives more access than User EXEC mode. It allows you to view system configurations, perform debugging, and run more advanced commands like backups and device restarts. However, configuration changes are not possible here. Common commands include:

- `show running-config`: displays the device's current configuration in RAM.

- `show startup-config`: displays the configuration stored in NVRAM.

- `reload`: restarts the device.

- `copy running-config startup-config`: saves the current running configuration to startup-config.

- `debug`: enables various debugging tools to troubleshoot issues.

To enter the Privileged Exec Mode, type the command ***enable*** from the User Exec Mode. You might be prompted for a password, depending on the device configuration. you would know that you are in the Privileged Exec Mode when you see "#" next to the router's name, e.g., Router#

To exit the Privileged EXEC Mode, type the command ***disable*** which will bring you back to the User EXEC Mode.

```
Router>enable
Router#?
Exec commands:
  <1-99>      Session number to resume
  auto        Exec level Automation
  clear       Reset functions
  clock       Manage the system clock
  configure   Enter configuration mode
  connect     Open a terminal connection
  copy        Copy from one file to another
  debug       Debugging functions (see also 'undebug')
  delete      Delete a file
  dir         List files on a filesystem
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
  erase       Erase a filesystem
  exit        Exit from the EXEC
  logout      Exit from the EXEC
  mkdir       Create new directory
  more        Display the contents of a file
  no          Disable debugging informations
  ping        Send echo messages
  reload      Halt and perform a cold restart

Router#disable
Router>
```

Figure 9: How to enter and exit the Privileged EXEC Mode along with its list of commands

5.3 Global Configuration Mode

This mode is used to make system-wide configuration changes. From here, you can configure interfaces, routing protocols, security settings, and more. It is the starting point for accessing more specific configuration sub-modes. From the Privileged EXEC Mode, you need to type *configure terminal* to access the Global Configuration Mode. You would know you are in the Global Configuration Mode when you see "(config)#" next to the device name. Common commands for this mode include:

- `hostname [name]`: changes the device's hostname.
- `enable secret [password]`: sets an encrypted password for privileged EXEC mode.
- `ip routing`: enables IP routing on a device.
- `interface [type/number]`: enters interface configuration mode.
- `router [protocol]`: enters routing protocol configuration mode.

You can return to the Privileged EXEC Mode by typing *exit*.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#?
Configure commands:
  aaa                 Authentication, Authorization and Accounting.
  access-list         Add an access list entry
  banner              Define a login banner
  bba-group           Configure BBA Group
  boot                Modify system boot parameters
  cdp                 Global CDP configuration subcommands
  class-map           Configure Class Map
  clock               Configure time-of-day clock
  config-register     Define the configuration register
  crypto              Encryption module
  default             Set a command to its defaults
  do                  To run exec commands in config mode
  dot11               IEEE 802.11 config commands
  enable              Modify enable password parameters
  end                 Exit from configure mode
  exit                Exit from configure mode
  flow                Global Flow configuration subcommands
  hostname            Set system's network name
  interface           Select an interface to configure
  ip                  Global IP configuration subcommands
  ipv6                Global IPv6 configuration commands

Router(config)#exit
Router#
```

Figure 10: How to enter and exit the Global Configuration Mode along with its commands

5.4 Interface Configuration Mode

This mode allows you to configure individual network interfaces on the device. You can set IP addresses, enable or disable the interface, configure duplex settings, and more. To enter this mode, you need first to be in the Global Configuration Mode. You then need to type interface [type/number], where type indicates the interface type (e.g., FastEthernet, GigabitEthernet, Serial, etc.) while the number specifies the number of the interface as a device can have multiple interfaces of the same type (e.g., GigabitEthernet 0/0). Common commands for this mode, include:

- `ip address [IP address] [subnet mask]`: assigns an IP address to the interface.
- `no shutdown`: switches on the interface (default is disabled).
- `shutdown`: disables the interface.
- `duplex [auto | full | half]`: configures duplex settings.
- `speed [auto | 10 | 100 | 1000]`: configures the speed of the interface.
- `description [text]`: adds a description to the interface.

You can type ***exit*** to return to Global Configuration Mode.

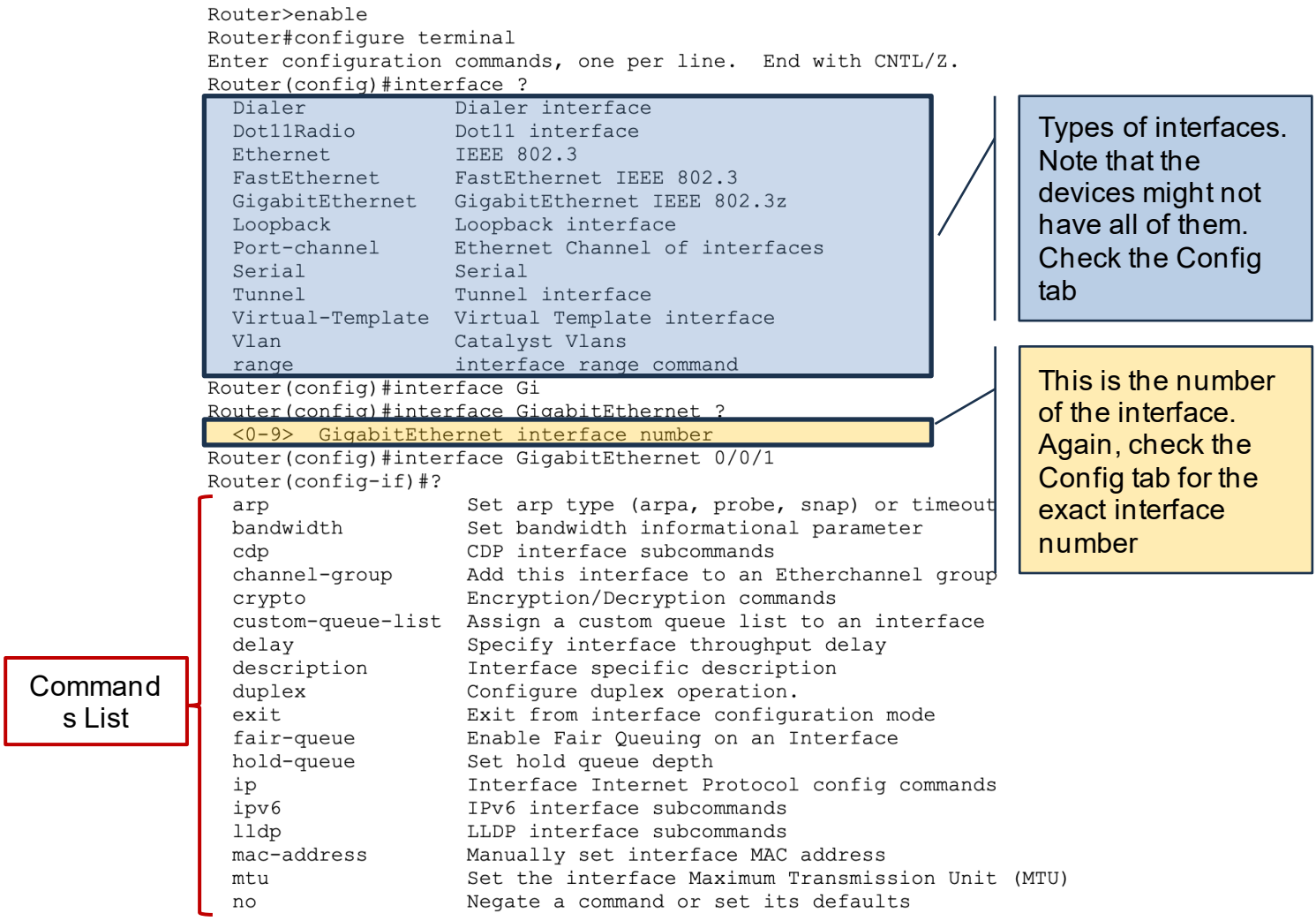


Figure 11: Interface Configuration Mode commands' list

5.5 Line Configuration Mode

This mode is similar to the Interface Configuration Mode and is used to configure settings for the terminal lines, including login security, access methods (such as SSH or Telnet), and idle timeouts. To enter this mode, type *line [type][number]* once you are in the Global Configuration mode. Common commands in this mode include:

- `password [password]`: Sets a password for the line.

- `login`: Enables password checking for the line.

- `exec-timeout [minutes] [seconds]`: Sets idle timeout before disconnecting the session.

- `transport input [telnet | ssh]`: Configures how the line is accessed remotely.

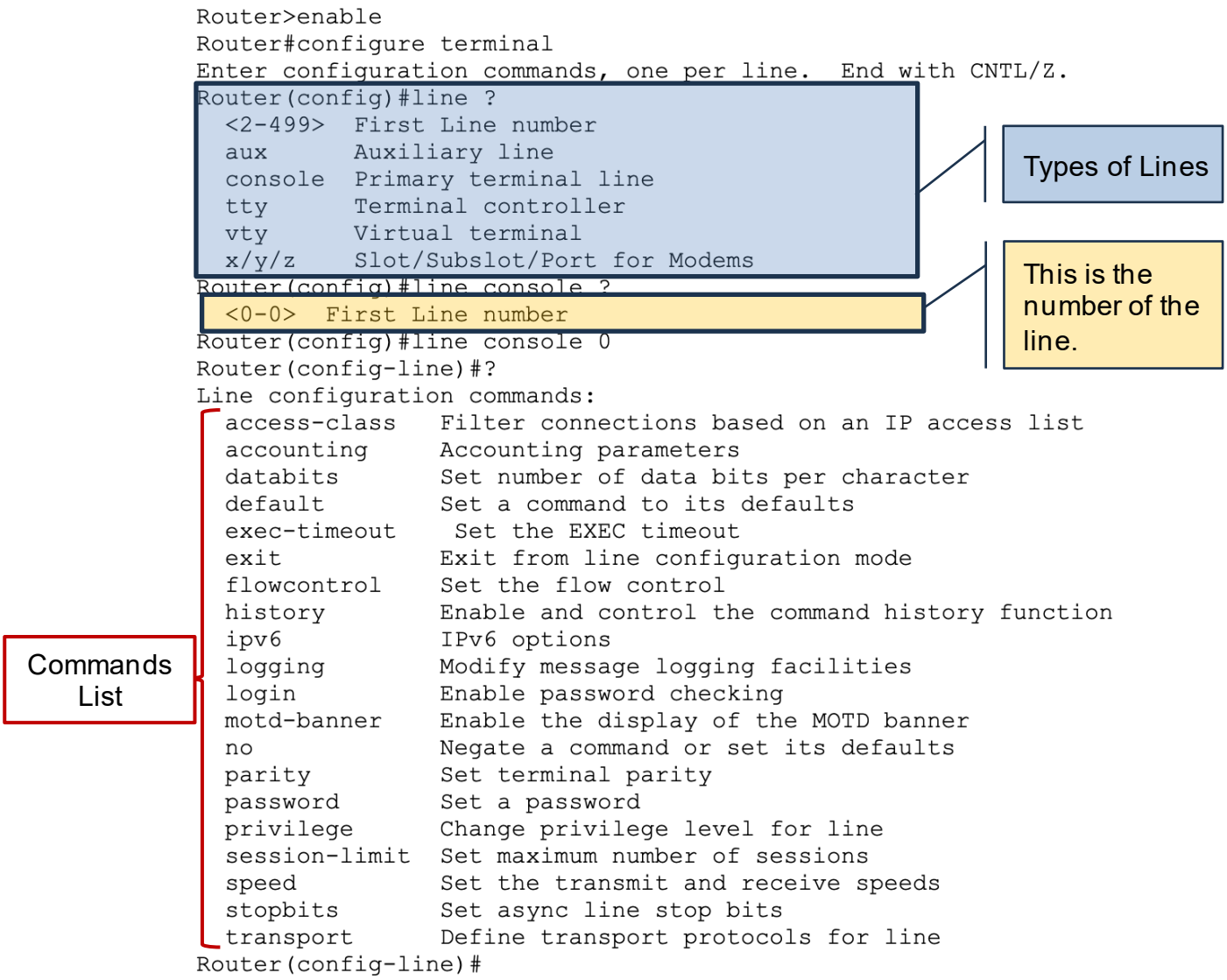


Figure 12: Line Configuration Mode and its list of commands

6. Running Configuration vs. Startup Configuration in Cisco Devices

When configuring Cisco devices such as routers and switches in Packet Tracer (or in real-world networking), the configuration changes you make are stored in two key places: the running configuration and the startup configuration. Understanding the difference between these two configurations is crucial for managing and saving network settings properly.

6.1 Running Configuration (running-config)

The running configuration holds the current, active configuration of the device. Any changes made to the device through CLI are immediately applied to the running-config, affecting the device's operation in real-time. Because it is stored in volatile memory (RAM), the running configuration is lost when the device is powered off or restarted.

6.2 Startup Configuration (startup-config)

The startup configuration contains the configuration that the device loads when it boots

up. This is a saved version of the running configuration, and it persists across device reboots or power cycles. Since it is stored in non-volatile memory (NVRAM), the startup configuration remains even when the device is powered off.

6.3 Working with Running and Startup Configurations

To ensure your changes are saved across reboots, you should always save your running configuration to the startup configuration. Here is a typical workflow when configuring a Cisco device:

1. Make configuration changes: enter Global Configuration Mode and make changes (e.g., configure interfaces, routing, security settings). These changes are stored in the running configuration immediately.
2. Verify your changes: use the show running-config command to verify that the changes have been applied correctly.
3. Save the configuration: to ensure these changes persist after a reboot, save the running configuration to the startup configuration using the following command: *copy running-config startup-config*
4. Reboot the device: when the device restarts, it will load the configuration stored in the startup-config, ensuring your settings are retained.